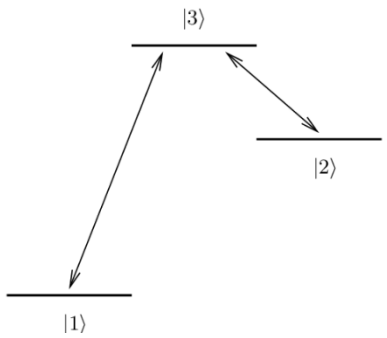
1-Dimensional 3-Level
Maxwell-Bloch equations



- **Atomic wave function**
  The general wave-function of the atom is given by:
  $$|\psi^m\rangle = C_1^m|1\rangle + C_2^m|2\rangle + C_3^m|3\rangle$$
- **Hamiltonian**
  The full Hamiltonian of the system is:
  $$\hat{H} = \hbar\omega_2|2\rangle\langle 2| + \hbar\omega_3|3\rangle\langle 3| + \hat{H}_{\text{int}}$$
  where the interaction Hamiltonian is typically written as:
  $$\hat{H}_{\text{int}} = -\vec{d}\cdot\vec{\mathcal{E}}$$
  with the dipole operator:
  $$\vec{d} = \vec{d}_{13}|1\rangle\langle 3| + \vec{d}_{23}|2\rangle\langle 3| + \text{c.c.}$$
  Where c.c. denotes complex conjugate.

Bloch equations

$$\hat{\rho}=\sum_m P_m\left|\psi^m\right\rangle\left\langle\psi^m\right|$$

$$\frac{\partial\hat{\rho}}{\partial t}=\frac{1}{i\hbar}\left[\hat{H},\hat{\rho}\right]\Rightarrow\begin{cases}\dot{\rho}_{11}=-\frac{i}{\hbar}\mathcal{E}(\rho_{13}d_{31}-\rho_{31}d_{13})\\ \dot{\rho}_{22}=-\frac{i}{\hbar}\mathcal{E}(\rho_{23}d_{32}-\rho_{32}d_{23})\\ \dot{\rho}_{33}=\frac{i}{\hbar}\mathcal{E}(\rho_{13}d_{31}-\rho_{31}d_{13})+\frac{i}{\hbar}\mathcal{E}(\rho_{23}d_{32}-\rho_{32}d_{23})\\ \dot{\rho}_{21}=-i\omega_2\rho_{21}-\frac{i}{\hbar}\mathcal{E}(\rho_{23}d_{31}-\rho_{31}d_{23})\\ \dot{\rho}_{31}=-i\omega_3\rho_{31}-\frac{i}{\hbar}\mathcal{E}d_{31}(\rho_{33}-\rho_{11})+\frac{i}{\hbar}\mathcal{E}d_{32}\rho_{21}\\ \dot{\rho}_{32}=-i(\omega_3-\omega_2)\rho_{32}-\frac{i}{\hbar}\mathcal{E}d_{32}(\rho_{33}-\rho_{22})+\frac{i}{\hbar}\mathcal{E}d_{31}\rho_{12}\end{cases}$$

$$\hat{\rho}=\hat{\rho}^\dagger\ \Rightarrow\ \rho_{ij}=\rho_{ji}^*$$

In general, the system can also include other states:

$$\left|\psi^m\right\rangle=C_1^m\left|1\right\rangle+C_2^m\left|2\right\rangle+C_3^m\left|3\right\rangle+C_{\text{os}}^m\left|\text{other states}\right\rangle$$

So:

$$\begin{aligned}\text{Tr}\left(\hat{\rho}\right)&=\sum_m P_m\left\langle\psi^m\left|\psi^m\right.\right\rangle\\&=\sum_m P_m\left|C_1^m\right|^2+\sum_m P_m\left|C_2^m\right|^2+\sum_m P_m\left|C_3^m\right|^2+\sum_m P_m\left|C_{\text{os}}^m\right|^2\\&=n_1(t)+n_2(t)+n_3(t)+n_{\text{os}}(t)\\&=n_{123}(t)+n_{\text{os}}(t)\end{aligned}$$

Where $n_{123}(t)$ is proportional to the probability that the atom is in states 1, 2, or 3.

Obviously:

$$\text{Tr}\left(\hat{\rho}\right)=n_{123}(0)+n_{\text{os}}(0)$$

In our convention, we take $n_{123}(0)=1$.

If we have N atoms in total, the number of atoms in these three states is $N_{123}(t)=\frac{n_{123}(t)}{\text{Tr}\left(\hat{\rho}\right)}N$, which initially is $N_{123}(0)=\frac{N}{\text{Tr}\left(\hat{\rho}\right)}$.

Maxwell equations (SI)

$$\nabla \times \mathcal{E} = -\frac{\partial \mathcal{B}}{\partial t} \quad , \quad \mathcal{D} = \epsilon_0 \mathcal{E} + \mathcal{P}$$

$$\nabla \times \mathcal{H} = \mathcal{J}_f + \frac{\partial \mathcal{D}}{\partial t} \quad , \quad \mathcal{H} = \frac{\mathcal{B}}{\mu_0} - \mathcal{M}$$

$$\nabla \times \mathcal{B} = \mu_0 \left(\mathcal{J}_f + \nabla \times \mathcal{M} + \frac{\partial \mathcal{P}}{\partial t} + \epsilon_0 \frac{\partial \mathcal{E}}{\partial t} \right)$$

$$\mathcal{J}_f = 0 \ , \ \mathcal{M} = 0 \quad \Rightarrow \quad \nabla \times \mathcal{B} = \frac{1}{c^2} \left(\frac{1}{\epsilon_0} \frac{\partial \mathcal{P}}{\partial t} + \frac{\partial \mathcal{E}}{\partial t} \right)$$

$$\nabla \times (\nabla \times \mathcal{E}) = -\frac{\partial}{\partial t} (\nabla \times \mathcal{B})$$

$$\nabla \times (\nabla \times \mathcal{E}) = -\frac{1}{c^2} \left(\frac{1}{\epsilon_0} \frac{\partial^2 \mathcal{P}}{\partial t^2} + \frac{\partial^2 \mathcal{E}}{\partial t^2} \right)$$

$$\nabla \times (\nabla \times \mathcal{E}) + \frac{1}{c^2} \frac{\partial^2 \mathcal{E}}{\partial t^2} = -\frac{1}{\epsilon_0 c^2} \frac{\partial^2 \mathcal{P}}{\partial t^2}$$

$$\nabla (\nabla \cdot \mathcal{E}) - \nabla^2 \mathcal{E} + \frac{1}{c^2} \frac{\partial^2 \mathcal{E}}{\partial t^2} = -\frac{1}{\epsilon_0 c^2} \frac{\partial^2 \mathcal{P}}{\partial t^2}$$

Assumptions:

- The system is one-dimensional (1D) $\Rightarrow \mathcal{E}(z,t)$.
- The electric field is perpendicular to its direction of propagation ($\mathcal{E} \perp \hat{z}$) $\Rightarrow \nabla \cdot \mathcal{E}(z,t) = 0$.
- The polarization is parallel to the electric field ($\mathcal{P} \parallel \mathcal{E}$).

$$\frac{\partial^2 \mathcal{E}}{\partial z^2} - \frac{1}{c^2} \frac{\partial^2 \mathcal{E}}{\partial t^2} = \frac{1}{\epsilon_0 c^2} \frac{\partial^2 \mathcal{P}}{\partial t^2} \quad ,$$

$$\mathcal{P} = \frac{N}{V} \frac{\mathrm{Tr}\left(\hat{d}\hat{\rho}\right)}{\mathrm{Tr}\left(\hat{\rho}\right)}$$

$$= \frac{N_{123}(0)}{V} \mathrm{Tr}\left(\hat{d}\hat{\rho}\right)$$

$$\mathcal{P} = n_0 \left(\rho_{31} d_{13} + d_{23} d_{23} + c.c. \right) \quad , \quad n_0 = \frac{N_{123}(0)}{V}$$

Maxwell-Bloch equations

The full coupled equations include the dynamics of the atomic populations and coherences (Bloch equations) and the electromagnetic field (Maxwell equations):

$$\dot{\rho}_{11} = -\frac{i}{\hbar}\mathcal{E}(\rho_{13}d_{31} - \rho_{31}d_{13})$$

$$\dot{\rho}_{22} = -\frac{i}{\hbar}\mathcal{E}(\rho_{23}d_{32} - \rho_{32}d_{23})$$

$$\dot{\rho}_{33} = \frac{i}{\hbar}\mathcal{E}(\rho_{13}d_{31} - \rho_{31}d_{13}) + \frac{i}{\hbar}\mathcal{E}(\rho_{23}d_{32} - \rho_{32}d_{23})$$

$$\dot{\rho}_{21} = -i\omega_2\rho_{21} - \frac{i}{\hbar}\mathcal{E}(\rho_{23}d_{31} - \rho_{31}d_{23})$$

$$\dot{\rho}_{31} = -i\omega_3\rho_{31} - \frac{i}{\hbar}\mathcal{E}d_{31}(\rho_{33} - \rho_{11}) + \frac{i}{\hbar}\mathcal{E}d_{32}\rho_{21}$$

$$\dot{\rho}_{32} = -i(\omega_3 - \omega_2)\rho_{32} - \frac{i}{\hbar}\mathcal{E}d_{32}(\rho_{33} - \rho_{22}) + \frac{i}{\hbar}\mathcal{E}d_{31}\rho_{12}$$

$$\frac{\partial^2\mathcal{E}}{\partial z^2} - \frac{1}{c^2}\frac{\partial^2\mathcal{E}}{\partial t^2} = \frac{1}{\epsilon_0 c^2}\frac{\partial^2\mathcal{P}}{\partial t^2}$$

$$\rho_{ij} = \rho_{ji}^* \quad , \quad d_{ij} = d_{ji}^* \quad , \quad \mathcal{P} = n_0\mathrm{Tr}\left(\hat{d}\hat{\rho}\right) = n_0(\rho_{31}d_{13} + d_{23}d_{23} + c.c.)$$

Approximation

- Rotating Wave Approximation (RWA)
  Neglect fast-oscillating terms of the form $e^{\pm 2i\omega_{ij}t}$.

- Slowly Varying Envelope Approximation (SVEA)
  The amplitudes are assumed to be slowly varying functions of their arguments:

	SVEA
$\mathcal{E}=\frac{1}{2}\{E_{31}e^{-i\varphi_{31}}+E_{32}e^{-i\varphi_{32}}+c.c.\}$	$\left\lvert\frac{\partial E_{ij}}{\partial t}\right\rvert \ll \omega_{ij}\lvert E_{ij}\rvert \quad , \quad \left\lvert\frac{\partial E_{ij}}{\partial z}\right\rvert \ll k_{ij}\lvert E_{ij}\rvert$
$\rho_{21}=\frac{1}{2}R_{21}e^{-i(\varphi_{31}-\varphi_{32})}$	
$\rho_{31}=\frac{1}{2}R_{31}e^{-i\varphi_{31}} \qquad \varphi_{ij}=\omega_{ij}t-k_{ij}z$	$\left\lvert\frac{\partial R_{ij}}{\partial t}\right\rvert \ll \omega_{ij}\lvert R_{ij}\rvert \quad , \quad \left\lvert\frac{\partial R_{ij}}{\partial z}\right\rvert \ll k_{ij}\lvert R_{ij}\rvert$
$\rho_{32}=\frac{1}{2}R_{32}e^{-i\varphi_{32}} \qquad \omega_{ij}=ck_{ij}$	

$$\mathcal{P}=\frac{n_0}{2}\left(d_{13}R_{31}e^{-i\varphi_{31}}+\rho_{32}R_{32}e^{-i\varphi_{32}}+c.c.\right)$$

$$\left(\frac{\partial^2}{\partial z^2}-\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)\frac{1}{2}\{E_{31}e^{-i\varphi_{31}}+E_{32}e^{-i\varphi_{32}}+c.c.\}=\left(\frac{1}{\epsilon_0 c^2}\frac{\partial^2}{\partial t^2}\right)\frac{n_0}{2}\left(d_{13}R_{31}e^{-i\varphi_{31}}+\rho_{32}R_{32}e^{-i\varphi_{32}}+c.c.\right)$$

$$\left(\frac{\partial^2}{\partial z^2}-\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)\{E_{31}e^{-i\varphi_{31}}\}=\frac{n_0}{\epsilon_0 c^2}\frac{\partial^2}{\partial t^2}\left(d_{13}R_{31}e^{-i\varphi_{31}}\right)$$

$$\begin{aligned}\left(\frac{\partial^2}{\partial z^2}-\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)\{E_{31}e^{-i\varphi_{31}}\}&=\frac{\partial^2}{\partial z^2}\left(E_{31}e^{-i\varphi_{31}}\right)-\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\left(E_{31}e^{-i\varphi_{31}}\right)\\&=\left(\frac{\partial^2 E_{31}}{\partial z^2}+2ik_{31}\frac{\partial E_{31}}{\partial z}-k_{31}^2E_{31}-\frac{1}{c^2}\left[\frac{\partial^2 E_{31}}{\partial t^2}-2i\omega_{31}\frac{\partial E_{31}}{\partial t}-\omega_{31}^2E_{31}\right]\right)e^{-i\varphi_{31}}\\&=\left(\underbrace{\left[\frac{\partial^2 E_{31}}{\partial z^2}-\frac{1}{c^2}\frac{\partial^2 E_{31}}{\partial t^2}\right]}_{\text{SVEA}=0}+2\left[ik_{31}\frac{\partial E_{31}}{\partial z}+\frac{i\omega_{31}}{c^2}\frac{\partial E_{31}}{\partial t}\right]-\underbrace{\left(k_{31}^2-\frac{\omega_{31}^2}{c^2}\right)}_{=0}E_{31}\right)e^{-i\varphi_{31}}\\&=2ik_{31}\left(\frac{\partial E_{31}}{\partial z}+\frac{1}{c}\frac{\partial E_{31}}{\partial t}\right)e^{-i\varphi_{31}}\end{aligned}$$

$$\frac{n_0}{\epsilon_0 c^2}\frac{\partial^2}{\partial t^2}\left(d_{13}R_{31}e^{-i\varphi_{31}}\right)\overset{\text{SVEA}}{=\!=}\frac{n_0 d_{13}}{\epsilon_0 c^2}\left(-\omega_{31}^2R_{31}e^{-i\varphi_{31}}\right)=-\frac{n_0 d_{13}}{\epsilon_0}\frac{\omega_{31}^2}{c^2}R_{31}e^{-i\varphi_{31}}=-\frac{n_0 d_{13}}{\epsilon_0}k_{31}^2R_{31}e^{-i\varphi_{31}}$$

$$2ik_{31}\left(\frac{\partial E_{31}}{\partial z}+\frac{1}{c}\frac{\partial E_{31}}{\partial t}\right)e^{-i\varphi_{31}}=-\frac{n_0 d_{13}}{\epsilon_0}k_{31}^2R_{31}e^{-i\varphi_{31}}$$

$$\begin{aligned}\frac{\partial E_{31}}{\partial z}+\frac{1}{c}\frac{\partial E_{31}}{\partial t}&=i\frac{n_0 d_{13}k_{31}}{2\epsilon_0}R_{31}\\\frac{\partial E_{32}}{\partial z}+\frac{1}{c}\frac{\partial E_{32}}{\partial t}&=i\frac{n_0 d_{23}k_{32}}{2\epsilon_0}R_{32}\end{aligned}\xrightarrow[t'=t-\frac{z}{c}]{z'=z}\begin{aligned}\frac{\partial E_{31}}{\partial z'}&=i\frac{n_0 d_{13}k_{31}}{2\epsilon_0}R_{31}\\\frac{\partial E_{32}}{\partial z'}&=i\frac{n_0 d_{23}k_{32}}{2\epsilon_0}R_{32}\end{aligned}$$

$$\dot{\rho}_{11} = -\frac{i}{\hbar}\mathcal{E}(\rho_{13}d_{31} - \rho_{31}d_{13})$$

$$= -\frac{i}{4\hbar}\{E_{31}e^{-i\varphi_{31}} + E_{32}e^{-i\varphi_{32}} + c.c.\}\,(R_{31}^{*}e^{i\varphi_{31}}d_{31} - R_{31}e^{-i\varphi_{31}}d_{13})$$

$$= -\frac{i}{4\hbar}\,(d_{31}E_{31}R_{31}^{*} - (d_{31}E_{31})^{*}R_{31})$$

$$= \frac{1}{4}\left[\left(-i\frac{d_{31}}{\hbar}E_{31}\right)R_{31}^{*} + \left(-i\frac{d_{31}}{\hbar}E_{31}\right)^{*}R_{31}\right]$$

$$= \frac{1}{2}\mathrm{Re}\left[\left(-i\frac{d_{31}}{\hbar}E_{31}\right)^{*}R_{31}\right]$$

$$\dot{\rho}_{22} = \frac{1}{2}\mathrm{Re}\left[\left(i\frac{d_{32}}{\hbar}E_{32}\right)^{*}R_{32}\right]$$

$$\dot{\rho}_{33} = -\frac{1}{2}\mathrm{Re}\left[\left(-i\frac{d_{31}}{\hbar}E_{31}\right)^{*}R_{31} + \left(i\frac{d_{32}}{\hbar}E_{32}\right)^{*}R_{32}\right]$$

$$\dot{\rho}_{21} = -i\omega_2\rho_{21} - \frac{i}{\hbar}\mathcal{E}(\rho_{23}d_{31} - \rho_{31}d_{23})$$

$$\frac{1}{2}\dot{R}_{21}e^{-i(\varphi_{31}-\varphi_{32})} - i\frac{\omega_{31}-\omega_{32}}{2}R_{21}e^{-i(\varphi_{31}-\varphi_{32})} = -i\frac{\omega_2}{2}R_{21}e^{-i(\varphi_{31}-\varphi_{32})}$$
$$- \frac{i}{4\hbar}\{E_{31}e^{-i\varphi_{31}} + E_{32}e^{-i\varphi_{32}} + c.c.\}\,(R_{32}^{*}e^{i\varphi_{32}}d_{31} - R_{31}e^{-i\varphi_{31}}d_{23})$$

$$\dot{R}_{21} = i\delta\omega_{21}R_{21} - \frac{i}{2\hbar}\,(d_{31}E_{31}R_{32}^{*} - (d_{32}E_{32})^{*}R_{31}) \quad , \quad \delta\omega_{21} = \omega_{31} - \omega_{32} - \omega_2$$

$$\dot{R}_{21} = i\delta\omega_{21}R_{21} + \frac{1}{2}\left[\left(-\frac{id_{31}}{\hbar}E_{31}\right)R_{32}^{*} + \left(-\frac{id_{32}}{\hbar}E_{32}\right)^{*}R_{31}\right]$$

$$\dot{\rho}_{31} = -i\omega_3\rho_{31} - \frac{i}{\hbar}\mathcal{E}d_{31}\,(\rho_{33} - \rho_{11}) + \frac{i}{\hbar}\mathcal{E}d_{32}\rho_{21}$$

$$\frac{1}{2}\dot{R}_{31}e^{-i\varphi_{31}} - i\frac{\omega_{31}}{2}R_{31}e^{-i\varphi_{31}} = -i\frac{\omega_3}{2}R_{31}e^{-i\varphi_{31}} - \frac{i}{2\hbar}\{E_{31}e^{-i\varphi_{31}} + E_{32}e^{-i\varphi_{32}} + c.c.\}d_{31}\,(\rho_{33} - \rho_{11})$$
$$+ \frac{i}{4\hbar}\{E_{31}e^{-i\varphi_{31}} + E_{32}e^{-i\varphi_{32}} + c.c.\}d_{32}R_{21}e^{-i(\varphi_{31}-\varphi_{32})}$$

$$\dot{R}_{31} = i\delta\omega_{31}R_{21} - \frac{i}{\hbar}d_{31}E_{31}\,(\rho_{33} - \rho_{11}) + \frac{i}{2\hbar}d_{32}E_{32}R_{21} \quad , \quad \delta\omega_{31} = \omega_{31} - \omega_3$$

$$\dot{R}_{31} = i\delta\omega_{31}R_{21} + \left(-\frac{id_{31}}{\hbar}E_{31}\right)(\rho_{33} - \rho_{11}) - \frac{1}{2}\left(-\frac{id_{32}}{\hbar}E_{32}\right)R_{21}$$

$$\dot{\rho}_{32} = -i(\omega_3 - \omega_2)\rho_{32} - \frac{i}{\hbar}\mathcal{E}d_{32}\,(\rho_{33} - \rho_{22}) + \frac{i}{\hbar}\mathcal{E}d_{31}\rho_{12}$$

$$\frac{1}{2}\dot{R}_{32}e^{-i\varphi_{32}} - i\frac{\omega_{32}}{2}R_{32}e^{-i\varphi_{32}} = -i\frac{\omega_3-\omega_2}{2}R_{32}e^{-i\varphi_{32}} - \frac{i}{2\hbar}\{E_{31}e^{-i\varphi_{31}} + E_{32}e^{-i\varphi_{32}} + c.c.\}d_{32}\,(\rho_{33} - \rho_{22})$$
$$+ \frac{i}{4\hbar}\{E_{31}e^{-i\varphi_{31}} + E_{32}e^{-i\varphi_{32}} + c.c.\}d_{31}R_{21}^{*}e^{i(\varphi_{31}-\varphi_{32})}$$

$$\dot{R}_{32} = i\delta\omega_{32}R_{32} - \frac{i}{\hbar}d_{32}E_{32}\,(\rho_{33} - \rho_{22}) + \frac{i}{2\hbar}d_{31}E_{31}R_{21}^{*} \quad , \quad \delta\omega_{31} = \omega_{32} - (\omega_3 - \omega_2)$$

$$\dot{R}_{32} = i\delta\omega_{32}R_{32} + \left(-\frac{id_{32}}{\hbar}E_{32}\right)(\rho_{33} - \rho_{22}) - \frac{1}{2}\left(-\frac{id_{31}}{\hbar}E_{31}\right)R_{21}^{*}$$

$$\dot{\rho}_{11} = \frac{1}{2}\mathrm{Re}\left[\left(-i\frac{d_{31}}{\hbar}E_{31}\right)^{*}R_{31}\right]$$

$$\dot{\rho}_{22} = \frac{1}{2}\mathrm{Re}\left[\left(-i\frac{d_{32}}{\hbar}E_{32}\right)^{*}R_{32}\right]$$

$$\dot{\rho}_{33} = -\frac{1}{2}\mathrm{Re}\left[\left(-i\frac{d_{31}}{\hbar}E_{31}\right)^{*}R_{31} + \left(-i\frac{d_{32}}{\hbar}E_{32}\right)^{*}R_{32}\right]$$

$$\dot{R}_{21} = i\delta\omega_{21}R_{21} + \frac{1}{2}\left[\left(-\frac{id_{31}}{\hbar}E_{31}\right)R_{32}^{*} + \left(-\frac{id_{32}}{\hbar}E_{32}\right)^{*}R_{31}\right] \qquad , \qquad \delta\omega_{21} = \omega_{31} - \omega_{32} - \omega_{2}$$

$$\dot{R}_{31} = i\delta\omega_{31}R_{21} + \left(-\frac{id_{31}}{\hbar}E_{31}\right)(\rho_{33} - \rho_{11}) - \frac{1}{2}\left(-\frac{id_{32}}{\hbar}E_{32}\right)R_{21} \qquad , \qquad \delta\omega_{31} = \omega_{31} - \omega_{3}$$

$$\dot{R}_{32} = i\delta\omega_{32}R_{32} + \left(-\frac{id_{32}}{\hbar}E_{32}\right)(\rho_{33} - \rho_{22}) - \frac{1}{2}\left(-\frac{id_{31}}{\hbar}E_{31}\right)R_{21}^{*} \qquad , \qquad \delta\omega_{31} = \omega_{32} - (\omega_{3} - \omega_{2})$$

$$\frac{\partial}{\partial z'}\left(-\frac{id_{31}}{\hbar}E_{31}\right) = \frac{n_{0}|d_{13}|^{2}k_{31}}{2\epsilon_{0}\hbar}R_{31}$$

$$\frac{\partial}{\partial z'}\left(-\frac{id_{32}}{\hbar}E_{32}\right) = \frac{n_{0}|d_{23}|^{2}k_{32}}{2\epsilon_{0}\hbar}R_{32}$$

Dimensionless variables

We introduce normalized variables to simplify the equations:

$$\tau = \frac{t'}{T_0} \quad , \quad \xi = \frac{z'}{L} \quad , \quad \tilde{E}_{ij} = -i\frac{T_0 d_{ij}}{\hbar} E_{ij} \quad , \quad n_i = \rho_{ii} \quad , \quad \dot{f} = \frac{\partial f}{\partial \tau} \quad , \quad \delta\omega'_{ij} = T_0 \delta\omega_{ij}$$

$$\dot{n}_1 = \frac{1}{2}\mathrm{Re}\left(\tilde{E}^*_{31} R_{31}\right)$$

$$\dot{n}_2 = \frac{1}{2}\mathrm{Re}\left(\tilde{E}^*_{32} R_{32}\right)$$

$$\dot{n}_3 = -\frac{1}{2}\mathrm{Re}\left(\tilde{E}^*_{31} R_{31} + \tilde{E}^*_{32} R_{32}\right)$$

$$\dot{R}_{21} = i\delta\omega'_{21} R_{21} + \frac{1}{2}\left(\tilde{E}_{31} R^*_{32} + \tilde{E}^*_{32} R_{31}\right)$$

$$\dot{R}_{31} = i\delta\omega'_{31} R_{31} + \tilde{E}_{31}(n_3 - n_1) - \frac{1}{2}\tilde{E}_{32} R_{21}$$

$$\dot{R}_{32} = i\delta\omega'_{32} R_{32} + \tilde{E}_{32}(n_3 - n_2) - \frac{1}{2}\tilde{E}_{31} R^*_{21}$$

$$\frac{\partial \tilde{E}_{31}}{\partial \xi} = \frac{n_0 L |d_{31}|^2 k_{31} T_0}{2\epsilon_0 \hbar} R_{31}$$

$$\frac{\partial \tilde{E}_{32}}{\partial \xi} = \frac{n_0 L |d_{32}|^2 k_{32} T_0}{2\epsilon_0 \hbar} R_{32}$$

$$T_0 = \frac{2\epsilon_0 \hbar}{n_0 L |d_{32}|^2 k_{32}} \quad \Rightarrow \quad \begin{aligned} \frac{\partial \tilde{E}_{31}}{\partial \xi} &= \eta R_{31} \\ \frac{\partial \tilde{E}_{32}}{\partial \xi} &= R_{32} \end{aligned} \quad , \quad \eta = \left|\frac{d_{31}}{d_{32}}\right|^2 \frac{k_{31}}{k_{32}} = \left|\frac{d_{31}}{d_{32}}\right|^2 \frac{\omega_{31}}{\omega_{32}} = d^2 \omega$$

Relaxation terms are added phenomenologically:

- Population decay: T_1.
- Coherence decay: T_2.

In this formulation, detuning terms are neglected (resonance assumed).

$\dot{n}_{31} = -\mathrm{Re}\left(\tilde{E}^*_{31} R_{31} + \frac{1}{2}\tilde{E}^*_{32} R_{32}\right) - \frac{n_{31} - n^0_{31}}{T_1^{31}} + \Lambda_n^{31}$	$\frac{\partial \tilde{E}_{31}}{\partial \xi} = \eta R_{31}$
$\dot{n}_{32} = -\mathrm{Re}\left(\frac{1}{2}\tilde{E}^*_{31} R_{31} + \tilde{E}^*_{32} R_{32}\right) - \frac{n_{32} - n^0_{32}}{T_1^{32}} + \Lambda_n^{32}$	$\frac{\partial \tilde{E}_{32}}{\partial \xi} = R_{32}$
$\dot{R}_{21} = \frac{1}{2}\left(\tilde{E}_{31} R^*_{32} + \tilde{E}^*_{32} R_{31}\right) - \frac{R_{21} - R^0_{21}}{T_2^{21}}$	
$\dot{R}_{31} = \tilde{E}_{31} n_{31} - \frac{1}{2}\tilde{E}_{32} R_{21} - \frac{R_{31} - R^0_{31}}{T_2^{31}}$	
$\dot{R}_{32} = \tilde{E}_{32} n_{32} - \frac{1}{2}\tilde{E}_{31} R^*_{21} - \frac{R_{32} - R^0_{32}}{T_2^{32}}$	